

Part B Chapter 13

Integrating Models¹

1 Introduction

As discussed in the first part of the course, a model is a representation of an object, phenomenon, system, theory, or idea (further referred to as process) in some form. Such a representation describes the most relevant (in principle not all) elements, relations and properties of the process considered, and is often used for further analysis of its characteristics and properties. A distinction can be made between a model of an existing process in reality and a model of a process that is still being designed, as often occurs in engineering disciplines. One way of expressing this distinction is as models for natural processes in contrast to models for artificial processes. In the latter case the model serves as a design description or design model (a kind of blueprint) that forms the basis to realise (or implement) the process being designed. Examples of such design models are architectural design models, mechanical design models, software architecture models, conceptual models for databases and information systems, agent models and knowledge models.

For practical application, often different types of models are needed in combination. By some form of integration such models are glued together to obtain a more complex, integrated model. Such an integration can concern:

- models for natural processes and models for artificial processes
- qualitative (logical) models and quantitative (numerical) models
- models for different subprocesses or domains
- models at different aggregation levels of a process

In the remainder of this course, the first type of integration is central. In particular, it will be addressed how within agent models (for example, used for the design of software agents), domain models of processes in reality (for example, models for human functioning) can be integrated. In addition, the integration of qualitative models and quantitative models will be briefly addressed.

2 Integration of Domain Models and Agent Models

Domain models are used to describe phenomena in the natural world such as for example:

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- the physical and biological environment,
- physiological and cognitive processes within humans,
- behavioural processes of a human in interaction with his or her environment
- social and organisational processes based on interaction between humans.

Agent models can be used to describe phenomena in the natural world (such as animals or humans and their behaviour and interaction), or for artificially created (engineered) processes such as, for example:

- software agents
- intelligent cars
- search bots and recommender systems on Internet
- smart homes
- robots

For this course, agent models will be used for both cases: to model (natural) human agents and to model (artificial) ambient agents. Here ‘ambient’ means in the immediate environment of humans that are supported by them, for example, on their cell phone. To model human functioning agent models will be addressed, for example, for humans with behaviour motivated by desires, and humans generating emotions and feelings. Moreover, for a number of applications it will be shown how domain models can be integrated within ambient agent models in order to give these ambient agents some understanding of the part of reality they are dealing with. Such understanding enables them to perform actions in a more informed, knowledgeable manner, and to show more human-like behaviour in interaction with humans.

3 Integration of qualitative models and quantitative models

Models for complex processes in reality (such as biological, neurological, psychological and social processes) are often based on quantitative, numerical methods. Such approaches often use numerical variables to describe aspects of the process and how they affect each other over time; for example, how the number of predators affects the number of preys. An advantage of such numerical approaches is that numerical approximation methods and software environments are available for simulation. Agent and knowledge modelling approaches are usually based on qualitative, logical languages. By means of logic-based approaches, declarative models can be specified using representation languages that are close to the natural language. An advantage of such declarative models is that they can be considered and analyzed at a high abstract level. Furthermore, automated support (e.g., dedicated editing and design tools) is provided for manipulation and design of models. A further advantage of such logical approaches is that they allow (automated) logical analysis of the relationships between different parts of a model, for example relationships between global properties of a multi-agent system as a whole, and local properties of the basic mechanisms within agents.

A (dynamic system) model, as it is considered here, is characterised by states and transitions between these states. One specific subclass is formed by models based on qualitative states and transitions. Another specific subclass is formed by numerical dynamic systems based on difference or differential equations. Models that incorporate both qualitative and quantitative elements are often called hybrid models. The LEADSTO modelling approach used as a vehicle in this course provides means to integrate the quantitative and qualitative aspects in such hybrid models.

In the remainder of this course, first Chapter 14 introduces the integration of domain models within agent models illustrated for a simple qualitative case concerning human behaviour motivated by desires. Next, Chapter 15 introduces integration of qualitative and quantitative aspects, for a case concerning biological populations and their management. In Chapter 16 and further integrative modelling for a number of themes will be addressed.